GAUSSIAN ELIMINATION

LINEAR ALGEBRA

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Reduced Row Echelon Form

Gaussian Elimination

REDUCED ROW ECHELON FORM

SOLUTIONS

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If this matrix represents a system of linear equations in the variables x, y, z, what are the solutions to the system?

PROPERTIES

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Properties of matrix in reduced row echelon form (RREF):

1. **Leading 1s**

   Each leading 1 is the only nonzero entry in its column

2. **Descending Order**

   The position of the leading 1 shifts progressively to the right (0's above it)

3. **Zero Rows**

   Any rows consisting entirely of zeros are at the bottom of the matrix

Which of these matrices are in **RREF**?

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

✓ RREF

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 5 & 2 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}$$

✓ RREF

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

✗ NOT

$$\begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}$$

✗ NOT

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

✗ NOT

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}$$

✗ NOT

GAUSSIAN ELIMINATION

Primary operations:

1. **Swapping Rows**

   Interchanging two rows to position a nonzero element

2. **Scaling Rows**

   Multiplying a row by a nonzero scalar to obtain a leading 1

3. **Row Addition/Subtraction**

   To create zeros

$$\begin{bmatrix} 2 & -4 & -2 & 12 \\ -1 & 2 & -1 & 0 \\ 5 & -5 & 3 & 6 \end{bmatrix}$$

Put the augmented matrix for the system of linear equations in **reduced row echelon form** and read-out the **solution**.

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cccc} a & b & c & \\ \end{array}$$

$$\begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \begin{array}{ll} r_1 & \leftarrow r_1/2 \\ r_2 & \leftarrow r_2/2 \\ r_3 & \leftarrow r_3/-4 \end{array}$$

$$\begin{array}{cccc} a & b & c & \\ \end{array}$$

$$\begin{bmatrix} 1 & -2 & -1 & 6 \\ 0 & -5 & -8 & 24 \\ 0 & 0 & 1 & -3 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

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Solution,

$\boxed{a = 3}$ *ans*

$\boxed{b = 0}$ *ans*

$\boxed{c = -3}$ *ans*

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SEATWORK